

Problem 9.16

If $i^{(8)} = 0.16$, calculate $d^{(1/2)}$.

Solution

The nominal interest rate convertible eight times per year is

$$i^{(8)} = 0.16.$$

Thus, the annual accumulation factor is

$$\left(1 + \frac{i^{(8)}}{8}\right)^8 = \left(1 + \frac{0.16}{8}\right)^8.$$

For the nominal discount rate $d^{(1/2)}$, the equivalent annual accumulation factor is

$$\left(1 - \frac{d^{(1/2)}}{1/2}\right)^{-1/2}.$$

Since the two rates are equivalent,

$$\left(1 + \frac{0.16}{8}\right)^8 = \left(1 - \frac{d^{(1/2)}}{1/2}\right)^{-1/2}.$$

Raising both sides to the power -2 gives

$$\left(1 - \frac{d^{(1/2)}}{1/2}\right) = \left(1 + \frac{0.16}{8}\right)^{-16}.$$

Therefore,

$$d^{(1/2)} = \frac{1}{2} \left[1 - \left(1 + \frac{0.16}{8}\right)^{-16} \right].$$

Substituting $1 + 0.16/8 = 1.02$,

$$d^{(1/2)} = \frac{1}{2} [1 - (1.02)^{-16}].$$

Finally,

$$d^{(1/2)} \approx 0.135777.$$

Hence,

$$\boxed{d^{(1/2)} \approx 13.5777\%}.$$